We implemented two baseline configurations: using only the PSG and using only the SAG. The proposed CHG serves as the complete model. The results are summarized in Figure 1 and Table 1.

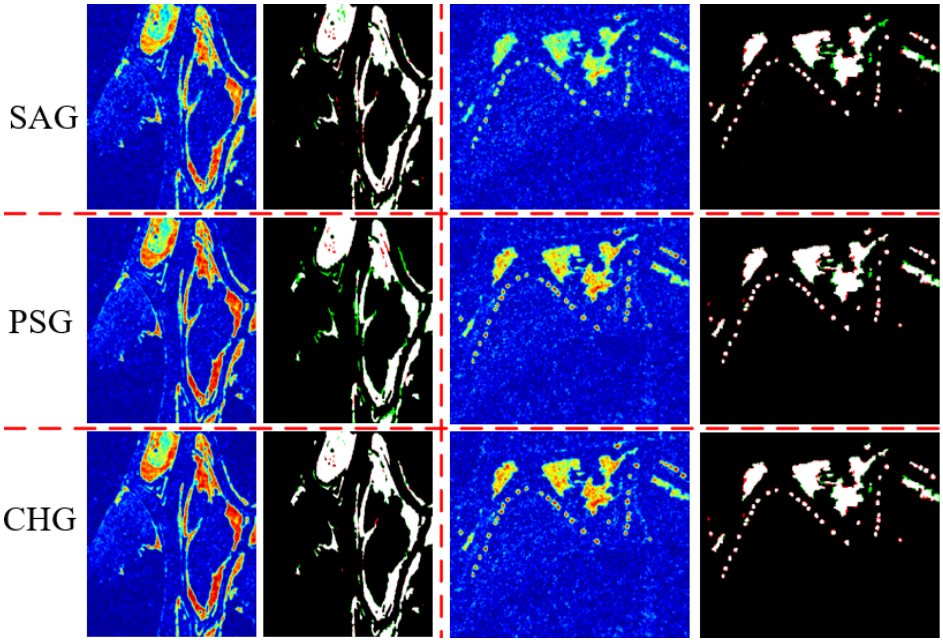

Figure 1. The qualitative results of ablation experiment.

Table 1. Quantitative results of ablation experiment.

Method	Ottawa					Lu'an I				
	FN	FP	OA	F1	κ	FN	FP	OA	F1	κ
PSG	2405	**405**	0.9723	0.9066	0.8905	**568**	1899	0.9852	0.8930	0.8851
SAG	**1064**	903	**0.9806**	**0.9384**	**0.9269**	1128	**1136**	**0.9864**	**0.8959**	**0.8886**
CHG	**671**	**577**	**0.9877**	**0.9610**	**0.9537**	**706**	**1287**	**0.9880**	**0.9107**	**0.9043**

PSG focuses on local patch-level features. As shown by the Ottawa dataset results, using PSG alone leads to a very high number of false negatives (FN = 2405) and a low Kappa coefficient (0.8905), indicating that without the macroscopic guidance of superpixels, patch-level analysis is susceptible to local noise interference and lacks regional contextual information. By coupling with SAG, CHG reduces FN to 671 and improves Kappa to 0.9537.

SAG captures regional homogeneity. While it outperforms PSG in noise suppression (e.g., κ = 0.9269 on Ottawa), it still lacks the fine details provided by patches. The complete CHG model achieves a κ improvement of 2.68% over SAG on the Ottawa dataset and 1.57% on the Lu'an I dataset.

The results consistently demonstrate that the coupled CHG architecture achieves the highest OA, F1 score, and κ across all datasets. This confirms that the coupling mechanism effectively integrates the noise robustness of superpixels with the detail preservation capability of patches, an advantage that cannot be attained by either single-scale graph (PSG or SAG) alone.